

Contents

01 - Getting Started	2
Prerequisites	3
Build Instructions	3
Project Initialization	3
Your First Program	3
Writing Comments	3
02 - Variables and Types	4
Constant Declaration with let	4
Variable Declaration with var	4
Type Annotations	4
Basic Types	4
String Operations	5
Escape Sequences	5
Reassignment Rules	5
Tuple Destructuring	6
03 - Operators	6
Arithmetic Operators	6
Comparison Operators	7
Logical Operators	7
Bitwise Operators	8
Compound Assignment Operators	8
Increment / Decrement Operators	8
Type Promotion Rules	9
Membership Operators	9
Control Flow	10
if / elif / else	10
while Loop	10
for Loop and range	10
break and continue	11
Nesting Example	12
Scope Rules	12
match	13
Functions	14
Basic Function Definition	14
Calling Functions	14
Recursive Functions	14
Function Overloading	15
Omitting the Return Type (Unit Type)	15
Structs and Enums	15
Defining Structs with record	15
Using Constructors	15
Field Access (Dot Notation)	16
Field Assignment	16
Structs as Function Parameters	16
Nested Structs	16
Enums	17
Collections	17

Tuples	17
Lists	18
Maps	20
Sets	21
Iterators	22
Advanced Features	23
Lambda Functions	23
Closures	24
Higher-Order Functions	24
Functions as Values	24
UFCS (Uniform Function Call Syntax)	24
Operator Overloading	25
Option Type	25
Concurrency Basics	26
F-String (String Interpolation)	26
Type Casting (as)	27
Enum with Associated Data (ADT)	27
Generic Enum	27
Result Type	28
Packages	28
from/import Syntax	28
Subdirectories (Dot-Separated Paths)	28
Directory Packages	29
Standard Library (std)	29
Search Path Priority	29
RY_PATH Environment Variable	29
Limitations	30
Design by Contract	30
Preconditions (require)	30
Postconditions (ensure)	30
Combining require and ensure	31
Struct Invariants (invariant)	31
Rules	31
Testing	32
Running Tests	32
Writing Tests	32
Matchers	32
Output Format	33
Mocking	33
Parameterized Tests	33
Property-Based Tests	34
Limitations	34

[English](#) | |

01 - Getting Started

Next tutorial -> [02 - Variables and Types](#)

Prerequisites

To build and run Ry, you need the following:

- **LLVM 21**
 - **CMake 3.20 or later**
 - **A C++17 compatible compiler** (GCC 7+ / Clang 5+, etc.)
-

Build Instructions

Run the following commands from the repository root:

```
cmake -B build -DLLVM_DIR=/usr/local/llvm/lib/cmake/llvm
cmake --build build
```

On a successful build, an executable named `build/ry` will be generated.

Project Initialization

You can create a new project with the `ry init` command:

```
mkdir my-project
cd my-project
ry init
```

This generates the following files and directories:

- `ry.toml` – Project configuration file
- `src/main.ry` – Entry point (with sample code)
- `test/` – Directory for test code

See [Project Management](#) for details.

Your First Program

Save the following content to a file named `hello.ry`:

```
print("Hello, World!")
```

Run it with the following command:

```
./build/ry hello.ry
```

Output:

```
Hello, World!
```

Writing Comments

Everything from `#` to the end of the line is treated as a comment.

```
# This is a comment
print("Hello") # End-of-line comments are also supported
```

Comments do not affect the behavior of the code.

Next tutorial -> [02 - Variables and Types](#)

[English](#) | |

02 - Variables and Types

<- [01 - Getting Started](#) / Next -> [03 - Operators](#)

Constant Declaration with let

The `let` keyword declares an immutable variable (constant). The type is automatically inferred from the value on the right-hand side. The value cannot be changed after declaration.

```
let x = 42           # Inferred as int
let y = 3.14         # Inferred as float
let flag = true      # Inferred as bool
let name = "Ry"      # Inferred as str
```

Variable Declaration with var

The `var` keyword declares a mutable variable. You can reassign a value of the same type after declaration.

```
var count = 0
count = count + 1  # OK: reassignment to the same type
```

Type Annotations

You can explicitly specify the type of a variable.

```
let x: int = 42
let rate: float = 0.5
let ok: bool = false
let msg: str = "hello"
```

A compile error occurs if the type annotation does not match the actual type of the value.

Basic Types

Type	Description	Literal Examples
<code>int</code>	64-bit integer	0, 42, -10
<code>byte</code>	Unsigned 8-bit integer (0-255)	<code>let b: byte = 42</code>
<code>float</code>	64-bit floating-point number	0.0, 3.14, -1.5
<code>bool</code>	Boolean	<code>true</code> , <code>false</code>
<code>str</code>	String	"hello", ""

String Operations

Various operations are available for strings.

```
let a = "Hello"
let b = "World"

# Concatenation
let c = a + ", " + b    # "Hello, World"

# Comparison (lexicographic order)
print(a == b)    # false
print(a != b)    # true
print(a < b)     # true ("H" < "W")

# Length
print(len(a))    # 5

# Substring checks
let s = "Hello, World!"
print(contains(s, "World"))    # true
print(starts_with(s, "Hello")) # true
print(ends_with(s, "!"))      # true
```

Escape Sequences

The following escape sequences can be used within strings.

Sequence	Meaning
<code>\n</code>	Newline
<code>\t</code>	Tab
<code>\\</code>	Backslash
<code>\"</code>	Double quote
<code>\0</code>	Null character

```
print("Hello\nWorld")    # Outputs on two lines
print("A\tB")            # Tab-separated
print("say \"hi\"")      # String containing double quotes
```

Reassignment Rules

Variables declared with `var` can be reassigned. However, the following restrictions apply:

```
var x = 10
x = 20    # OK: reassignment to the same type
# x = "text" # Error: reassignment with a different type is not allowed
```

`let` variables cannot be reassigned.

```
let N = 5
# N = 6 # Error: reassignment to a let variable is not allowed
```

Redeclaring a variable with the same name is also not allowed.

```
let x = 1
# let x = 2 # Error: redeclaring the same name is not allowed
```

Tuple Destructuring

You can unpack a tuple into multiple variables in a single declaration using `let` or `var`.

```
let a, b = (10, 20)
print(a)    # 10
print(b)    # 20
```

Wildcard

Use `_` to ignore a position.

```
let x, _ = (1, 2)    # only x is bound; 2 is discarded
print(x)             # 1
```

Mutable Destructuring

Use `var` to declare mutable variables.

```
var a, b = (10, 20)
a = 99
print(a)    # 99
```

Rules

- The number of variables on the left must match the number of elements in the tuple.
 - Each variable follows the same `let`/`var` rules as a regular declaration.
 - Nested tuple destructuring is not supported.
-

[<- 01 - Getting Started](#) / [Next -> 03 - Operators](#)

[English](#) | [|](#)

03 - Operators

[<- 02 - Variables and Types](#) / [Next -> 04 - Control Flow](#)

Arithmetic Operators

Operator	Description	Example	Result
+	Addition	3 + 2	5
-	Subtraction	3 - 2	1
*	Multiplication / string repetition	3 * 2	6
/	Division (always float)	7 / 2	3.5
//	Integer division (always int)	7 // 2	3
%	Modulo	7 % 3	1
**	Exponentiation (always float)	2 ** 10	1024.0

```

let a = 10
let b = 3

print(a + b)    # 13
print(a - b)    # 7
print(a * b)    # 30
print(a / b)    # 3.3333... (float)
print(a // b)   # 3 (int)
print(a % b)    # 1
print(2 ** 8)   # 256.0 (float)

```

Comparison Operators

All comparison operators return a bool value.

Operator	Description	Example
==	Equal to	a == b
!=	Not equal to	a != b
<	Less than	a < b
<=	Less than or equal to	a <= b
>	Greater than	a > b
>=	Greater than or equal to	a >= b

```

let x = 5
let y = 10

print(x == y)    # false
print(x != y)    # true
print(x < y)     # true
print(x <= y)    # true
print(x > y)     # false
print(x >= y)    # false

```

Comparison operators also work on strings (lexicographic comparison).

```

print("abc" == "abc")    # true
print("abc" < "abd")     # true
print("b" > "a")         # true

```

Logical Operators

Operator	Description	Example
and	Logical AND	a and b
or	Logical OR	a or b
not	Logical NOT	not a

```

let t = true
let f = false

print(t and f)    # false

```

```
print(t or f)    # true
print(not t)     # false
print(not f)     # true
```

Bitwise Operators

Bitwise operators can only be used with the `int` type.

Operator	Description	Example
<code>&</code>	Bitwise AND	<code>5 & 3 -> 1</code>
<code> </code>	Bitwise OR	<code>5 3 -> 7</code>
<code>^</code>	Bitwise XOR	<code>5 ^ 3 -> 6</code>
<code>~</code>	Bitwise NOT (unary)	<code>~5 -> -6</code>
<code><<</code>	Left shift	<code>1 << 3 -> 8</code>
<code>>></code>	Right shift (arithmetic)	<code>8 >> 2 -> 2</code>
<code>>>></code>	Logical right shift	<code>-1 >>> 1 -> 9223372036854775807</code>

```
let a = 0b1010    # 10
let b = 0b1100    # 12

print(a & b)      # 8 (0b1000)
print(a | b)      # 14 (0b1110)
print(a ^ b)      # 6 (0b0110)
print(~a)         # -11
print(1 << 4)     # 16
print(32 >> 2)    # 8
```

Compound Assignment Operators

Shorthand notation for updating the value of a variable.

Operator	Description	Equivalent Expression
<code>+=</code>	Addition assignment	<code>x = x + n</code>
<code>-=</code>	Subtraction assignment	<code>x = x - n</code>
<code>*=</code>	Multiplication assignment	<code>x = x * n</code>
<code>/=</code>	Division assignment	<code>x = x / n</code>
<code>%=</code>	Modulo assignment	<code>x = x % n</code>

```
let x = 10
x += 5    # x == 15
x -= 3    # x == 12
x *= 2    # x == 24
x /= 4    # x == 6.0 (becomes float)
```

Increment / Decrement Operators

Shorthand for incrementing or decrementing a variable by 1.

Operator	Description	Equivalent Expression
<code>x++</code>	Increment by 1	<code>x = x + 1</code>
<code>x--</code>	Decrement by 1	<code>x = x - 1</code>

```
var count = 0
count++    # count == 1
count++    # count == 2
count--    # count == 1
```

Note: These are statement-only operators. They cannot be used inside expressions.

Type Promotion Rules

The following describes the behavior when `int` and `float` are mixed in operations.

```
# + - * produce float if either operand is float
print(1 + 2)      # 3 (int)
print(1 + 2.0)    # 3.0 (float)
print(1.0 + 2)    # 3.0 (float)

# / always produces float
print(4 / 2)      # 2.0 (float)

# // always produces int
print(7 // 2)     # 3 (int)
print(7.0 // 2)   # 3 (int)

# ** always produces float
print(2 ** 3)     # 8.0 (float)

# % produces int if both operands are int, float if either is float
print(7 % 3)      # 1 (int)
print(7.5 % 2)    # 1.5 (float)

# + concatenates strings if both operands are str
print("foo" + "bar") # "foobar"

# * repeats a string if one operand is str and the other is int
print("ab" * 3)    # "ababab"
print(3 * "ab")    # "ababab"
```

Membership Operators

Operator	Description	Example
<code>in</code>	Membership test	<code>2 in {1, 2, 3} -> true</code>
<code>not in</code>	Negated membership test	<code>4 not in {1, 2, 3} -> true</code>

```
let s = {1, 2, 3}
print(2 in s)      # true
print(4 not in s)  # true
```

[<- 02 - Variables and Types](#) / [Next -> 04 - Control Flow](#)

[English](#) | |

Control Flow

[<- Prev: Operators](#) | [Next: Functions ->](#)

if / elif / else

Use `if` to branch execution based on conditions.

```
let x = 10
```

```
if x > 0:
    print(x)
elif x == 0:
    print(0)
else:
    print(-1)
```

- `elif` and `else` are optional.
- Conditions are not limited to `bool` values. For `int`, 0 is treated as false and non-0 as true.
- `if` statements can be nested.

```
let a = 5
let b = 3
```

```
if a > 0:
    if b > 0:
        print(a + b)    # 8
```

while Loop

Repeatedly executes a block as long as the condition is true.

```
let i = 3
while i > 0:
    print(i)
    i = i - 1
# 3
# 2
# 1
```

for Loop and range

You can iterate over a list or using `range`.

```
for x in [1, 2, 3]:
    print(x)
# 1
```

```
# 2
# 3
```

`range(n)` generates integers from 0 to `n - 1`.

```
for i in range(5):
    print(i)
# 0
# 1
# 2
# 3
# 4
```

`range(start, end)` generates integers from `start` to `end - 1`.

```
for i in range(2, 5):
    print(i)
# 2
# 3
# 4
```

The `..` range operator creates an inclusive range: `1 .. 3` produces `[1, 2, 3]`.

```
for i in 1 .. 3:
    print(i)
# 1
# 2
# 3
```

You can iterate over map key-value pairs with `for k, v in map:`

```
let m = {"x": 10, "y": 20}
for k, v in m:
    print(k)
    print(v)
```

break and continue

`break` immediately exits the loop. `continue` skips the current iteration and proceeds to the next one.

```
for i in range(10):
    if i == 5:
        break
    if i % 2 == 0:
        continue
    print(i)
# 1
# 3
```

They can also be used with `while` loops.

```
let n = 0
while true:
    n = n + 1
    if n % 2 == 0:
        continue
    if n > 7:
        break
```

```
    print(n)
# 1
# 3
# 5
# 7
```

Note: In nested loops, `break` / `continue` only affect the innermost loop. Using them outside a loop results in a compile error.

Nesting Example

`for` and `while` loops can be nested.

```
for i in range(1, 4):
    for j in range(1, 4):
        if j == 2:
            continue
        print(i * 10 + j)
# 11
# 13
# 21
# 23
# 31
# 33
```

Scope Rules

Control flow blocks have their own scope.

Block Scope

Variables declared inside a block cannot be referenced from outside the block.

```
if true:
    let inner = 42
# Referencing inner here causes a compile error
```

Referencing and Reassigning Outer Variables

You can reference and reassign outer variables from within a block.

```
let count = 0
for i in range(5):
    count = count + i
print(count)    # 10
```

Shadowing

If you declare a variable with the same name as an outer variable inside a block, the new variable is used within that block (shadowing). The outer variable remains unchanged.

```
let x = 1
if true:
    let x = 99
```

```
    print(x)    # 99
print(x)        # 1
```

match

`match` is a construct for branching based on a value. It can safely handle enums and Options.

```
enum Color:
    Red
    Green
    Blue

let c = Color::Green
match c:
    case Color::Red:
        print("red")
    case Color::Green:
        print("green")
    case Color::Blue:
        print("blue")
# green
```

Option Matching

Use `match` to safely handle both the `Some` and `None` cases.

```
let x: Option<int> = Some(42)
match x:
    case Some(v):
        print(v)
    case None:
        print("nothing")
# 42
```

Wildcards and Literals

`_` is a wildcard pattern that matches anything. You can also match against literal values (numbers, strings, booleans).

```
let n = 5
match n:
    case 0:
        print("zero")
    case 1:
        print("one")
    case _:
        print("other")
# other
```

Guard Clauses

You can add guard conditions with `if`.

```
match n:
    case x if x > 0:
        print("positive")
```

```
case x if x < 0:
    print("negative")
case _:
    print("zero")
```

Note: `match` must be exhaustive. For enums, all variants must be covered. For Options, both `Some` and `None` are required. For literals, a `_` wildcard is needed.

[<- Prev: Operators](#) | [Next: Functions ->](#)

[English](#) | |

Functions

[<- Prev: Control Flow](#) | [Next: Structs ->](#)

Basic Function Definition

Functions are defined with the `fn` keyword. Parameter type declarations are required and use the `name: type` format. The return type is specified after `->`.

```
fn add(a: int, b: int) -> int:
    return a + b
```

- Parameter type declarations are required.
 - The return type is specified after `->`.
 - Use `return` to return a value.
-

Calling Functions

Call a defined function by its name with arguments.

```
fn multiply(x: int, y: int) -> int:
    return x * y

let result = multiply(3, 4)
print(result)    # 12
```

Recursive Functions

Functions can call themselves (recursion).

```
fn factorial(n: int) -> int:
    if n <= 1:
        return 1
    return n * factorial(n - 1)

print(factorial(5))    # 120
print(factorial(0))    # 1
```

Function Overloading

You can define multiple functions with the same name but different parameter counts or types.

```
fn add(a: int, b: int) -> int:
    return a + b

fn add(a: float, b: float) -> float:
    return a + b

print(add(1, 2))      # 3
print(add(1.5, 2.5))  # 4
```

The appropriate function is automatically selected based on the argument types at the call site.

Note: Defining functions with identical parameter types but different return types causes a compile error.

Omitting the Return Type (Unit Type)

If a function does not need to return a value, you can omit `->`. In this case, the function returns the Unit type.

```
fn greet():
    print(42)

greet()    # 42
```

This is the simplest form of a function with no parameters and no return value.

[<- Prev: Control Flow](#) | [Next: Structs ->](#)

[English](#) | |

Structs and Enums

[<- Prev: Functions](#) | [Next: Collections ->](#)

Defining Structs with record

Structs are defined with the `record` keyword. Each field is described in the `name: type` format.

```
record Point:
    x: int
    y: int
```

Structs are value types allocated on the stack.

Using Constructors

Create an instance by calling the struct name as a function. Arguments are specified in the order the fields are defined.

```
let p = Point(10, 20)
```

Field Access (Dot Notation)

Fields are accessed using dot notation.

```
let p = Point(10, 20)
print(p.x)    # 10
print(p.y)    # 20
```

Note: Passing a struct directly to `print` causes an error. Pass individual fields instead.

Field Assignment

Fields of variables declared with `var` can be reassigned.

```
var p = Point(10, 20)
p.x = 100
print(p.x)    # 100
```

Note: Assigning to fields of a `let` variable causes a compile error.

Structs as Function Parameters

Structs can be passed as function arguments.

```
record Point:
  x: int
  y: int

fn distance_x(a: Point, b: Point) -> int:
  return a.x - b.x

let p1 = Point(10, 3)
let p2 = Point(4, 7)
print(distance_x(p1, p2))    # 6
```

Nested Structs

A struct's field can be another struct.

```
record Point:
  x: int
  y: int

record Line:
  start: Point
  end: Point

let line = Line(Point(0, 0), Point(10, 5))
print(line.start.x)    # 0
print(line.end.x)      # 10
```

You can access nested fields by chaining dot notation.

Enums

Enums are defined with the `enum` keyword. Each variant is treated as a named constant.

Definition

```
enum Color:
  Red
  Green
  Blue
```

Usage

Variants are accessed with `::`.

```
let c = Color::Red
print(c)    # Red
```

Comparison

Variants can be compared using `==` and `!=`.

```
if c == Color::Red:
  print("red!")
elif c == Color::Green:
  print("green!")
else:
  print("blue!")
```

Function Parameters

Enum names can be used as function parameter types.

```
fn describe(c: Color) -> str:
  if c == Color::Red:
    return "warm"
  return "cool"
```

[<- Prev: Functions](#) | [Next: Collections ->](#)

[English](#) | |

Collections

[<- Prev: Structs](#) | [Next: Advanced Features ->](#)

Ry has four collection types: **Tuples**, **Lists**, **Maps**, and **Sets**.

Tuples

A tuple is an immutable data structure that groups multiple values together. It can hold elements of different types.

Creation

```
let t = (1, 3.14)
```

Type Annotation

```
let t: (int, float) = (1, 3.14)
```

Element Access

Elements are accessed by index using `.0`, `.1`, etc.

```
let t = (1, 3.14)
print(t.0)    # 1
print(t.1)    # 3.14
```

As Function Return Values

Tuples are useful when you want to return multiple values.

```
fn swap(a: int, b: int) -> (int, int):
    return (b, a)
```

```
let result = swap(1, 2)
print(result.0) # 2
print(result.1) # 1
```

Limitations

- Accessing an out-of-bounds index (e.g., `.2` on a tuple with 2 elements) causes a compile error.
 - Passing a tuple directly to `print` causes an error. Pass each element individually.
-

Lists

A list is a variable-length data structure containing elements of the same type.

Creation

```
let xs = [1, 2, 3]
```

Type Annotation

```
let xs: List<int> = [1, 2, 3]
```

Index Access

```
print(xs[0])    # 1
```

```
let i = 1
print(xs[i])    # 2
```

Index Assignment

```
xs[0] = 99
```

len

```
print(len(xs))    # 3
```

print

```
print(xs)         # [1, 2, 3]
```

Iteration with for

```
for x in xs:
    print(x)
```

Function Parameters

```
fn first(xs: List<int>) -> int:
    return xs[0]
```

filter, map, sort

Lists support filter, map, and sort operations. These return new lists without modifying the original.

```
let xs = [1, 2, 3, 4, 5]
```

```
# filter: keep elements matching a condition
```

```
let evens = xs.filter(fn(x: int): x > 3)
print(evens)    # [4, 5]
```

```
# map: transform each element
```

```
let doubled = xs.map(fn(x: int): x * 2)
print(doubled)  # [2, 4, 6, 8, 10]
```

```
# sort: sort in ascending order (default)
```

```
let sorted = [3, 1, 2].sort()
print(sorted)   # [1, 2, 3]
```

```
# Chaining
```

```
let result = xs.filter(fn(x: int): x > 1).map(fn(x: int): x * 10).sort()
print(result)  # [20, 30, 40, 50]
```

reduce, fold

reduce accumulates a list into a single value starting from the first element. fold does the same but with an explicit initial value.

```
let xs = [1, 2, 3, 4, 5]
```

```
# reduce: start from first element
```

```
let total = reduce(xs, fn(a: int, b: int): a + b)
print(total)    # 15
```

```
# fold: provide an explicit initial value
```

```
let total2 = fold(xs, 0, fn(a: int, b: int): a + b)
print(total2)   # 15
```

any, all

`any` returns `true` if at least one element satisfies the predicate. `all` returns `true` if every element does.

```
let xs = [1, 2, 3, 4, 5]

print(any(xs, fn(x: int): x > 4))  # true
print(any(xs, fn(x: int): x > 9))  # false

print(all(xs, fn(x: int): x > 0))  # true
print(all(xs, fn(x: int): x > 3))  # false
```

sum, min, max

```
let xs = [3, 1, 4, 1, 5]
print(sum(xs))  # 14
print(min(xs))  # 1
print(max(xs))  # 5
```

first, last, is_empty

```
let xs = [10, 20, 30]
print(first(xs))  # 10
print(last(xs))   # 30
print(is_empty(xs))  # false
```

enumerate, zip

`enumerate` pairs each element with its index. `zip` combines two lists element-by-element.

```
let xs = [10, 20, 30]
let indexed = enumerate(xs)
# [(0, 10), (1, 20), (2, 30)]
for p in indexed:
    print(p.0)
    print(p.1)

let ys = ["a", "b", "c"]
let zipped = zip(xs, ys)
# [(10, "a"), (20, "b"), (30, "c")]
```

Limitations

- All elements must be of the same type. Mixing different types is not allowed.
 - An empty list `[]` causes an error.
 - Out-of-bounds access results in a runtime error (`exit(1)`).
 - Supported element types are `int`, `float`, `bool`, and `str`.
-

Maps

A map is an associative array that manages key-value pairs.

Creation

```
let m = {"a": 1, "b": 2}
```

Type Annotation

```
let m: Map<str, int> = {"a": 1, "b": 2}
```

Key Access

```
print(m["a"])    # 1
```

Insertion / Update

Assigning to a new key inserts it; assigning to an existing key updates it.

```
m["c"] = 3        # Insert new entry
m["a"] = 99       # Update existing entry
```

len

```
print(len(m))    # 3
```

print

```
print(m)         # {a: 99, b: 2, c: 3}
```

has_key

Checks whether a key exists.

```
print(m.has_key("a"))    # true
```

keys, values

`keys` returns a list of all keys. `values` returns a list of all values.

```
let m = {"a": 1, "b": 2, "c": 3}
print(keys(m))      # ["a", "b", "c"]
print(values(m))    # [1, 2, 3]
```

Function Parameters

```
fn get_val(m: Map<str, int>, k: str) -> int:
    return m[k]
```

Limitations

- All keys must be of the same type, and all values must be of the same type.
 - An empty map requires a type annotation.
 - Accessing a nonexistent key results in a runtime error (`exit(1)`).
-

Sets

A set is a collection that holds elements of the same type without duplicates.

Creation

```
let s = {1, 2, 3}
```

Type Annotation

```
let s: Set<int> = {1, 2, 3}
```

in Operator

Use the `in` operator to check if an element is in the set.

```
print(2 in s)    # true
print(5 in s)    # false
```

add / remove

```
s.add(4)         # Add element
s.remove(1)      # Remove element
s.add(2)         # Ignored since it already exists
```

len / print

```
print(len(s))    # 3
print(s)         # {2, 3, 4}
```

Iteration with for

```
for x in s:
    print(x)
```

Empty Set

An empty set requires a type annotation.

```
let empty: Set<int> = {}
```

Limitations

- All elements must be of the same type.
 - Supported element types are `int`, `float`, `bool`, and `str`.
-

Iterators

Iterators provide a **lazy** way to process collections. Instead of creating intermediate lists at each step, iterators process elements one at a time through a pipeline.

Creating and Consuming

```
let xs = [1, 2, 3]
let ys = xs.iter().to_list()    # [1, 2, 3]
```

Chaining Operations

You can chain `filter`, `map`, and `take` to build pipelines:

```
let result = [1, 2, 3, 4, 5]
    .iter()
    .filter(fn(x: int): x > 2)
    .map(fn(x: int): x * 2)
    .take(2)
```

```
    .to_list()
print(result)    # [6, 8]
```

Manual Iteration with next()

```
let it = [10, 20].iter()
print(it.next())    # Some(10)
print(it.next())    # Some(20)
print(it.next())    # None
```

For Loops

Iterators work directly in for loops:

```
for x in [1, 2, 3].iter().filter(fn(x: int): x > 1):
    print(x)    # 2, 3
```

Maps produce tuple elements:

```
for k, v in {"a": 1, "b": 2}.iter():
    print(k)
```

[<- Prev: Structs](#) | [Next: Advanced Features ->](#)

[English](#) | |

Advanced Features

[<- Prev: Collections](#) | [Next: Packages ->](#)

Lambda Functions

Lambda functions are a syntax for writing functions as expressions. They use the form `fn(parameters): expression`. The return type is automatically inferred.

Single-Expression Lambda

```
let double = fn(x: int): x * 2
print(double(5))    # 10
```

```
let add = fn(a: int, b: int): a + b
print(add(3, 4))    # 7
```

No-Parameter Lambda

```
let answer = fn(): 42
print(answer())    # 42
```

Multi-Line Lambda

You can write multiple statements by adding a newline after `:` and indenting.

```
let abs = fn(x: int):
    if x < 0:
        return -x
```

```
    return x

print(abs(-5))  # 5
print(abs(3))   # 3
```

Closures

Lambda functions can capture variables from the scope in which they are defined.

```
let offset = 10
let add_offset = fn(x: int): x + offset
print(add_offset(5))  # 15
```

Higher-Order Functions

You can define functions that take other functions as arguments. Function types are written as `fn(parameter_types) -> return_type`.

```
fn apply(f: fn(int) -> int, x: int) -> int:
    return f(x)

let double = fn(x: int): x * 2
print(apply(double, 3))           # 6
print(apply(fn(n: int): n + 1, 10)) # 11
```

Functions as Values

Named functions can also be bound to variables or passed as arguments.

```
fn square(x: int) -> int:
    return x * x

fn apply(f: fn(int) -> int, x: int) -> int:
    return f(x)

# Pass a named function as an argument
print(apply(square, 4))  # 16

# Bind to a variable
let sq = square
print(sq(5))  # 25
```

UFCS (Uniform Function Call Syntax)

With UFCS, you can write `f(a, b)` as `a.f(b)`. This enables method-chaining-style notation.

```
fn add(a: int, b: int) -> int:
    return a + b

let x = 1
print(x.add(2))  # add(x, 2) -> 3
```


Chained Calls

```
fn double(n: int) -> int:
    return n * 2

print(x.add(2).double())    # double(add(x, 2)) -> 6
```

Operator Overloading

You can define operators for custom types using the `fn operator` syntax.

Binary Operators

Takes two parameters.

```
record Vec2:
    x: int
    y: int

fn operator+(a: Vec2, b: Vec2) -> Vec2:
    return Vec2(a.x + b.x, a.y + b.y)

fn operator==(a: Vec2, b: Vec2) -> bool:
    return a.x == b.x and a.y == b.y

let v1 = Vec2(1, 2)
let v2 = Vec2(3, 4)
let v3 = v1 + v2
print(v3.x)        # 4
print(v1 == v2)    # false
```

Unary Operators

Takes one parameter.

```
fn operator-(v: Vec2) -> Vec2:
    return Vec2(-v.x, -v.y)
```

Supported Operators

Category	Operators
Arithmetic	+, -, *, /, %, **, //
Comparison	==, !=, <, <=, >, >=
Bitwise	&, \, ^, ~, <<, >>
Logical	and, or, not

Option Type

A type that represents whether a value exists or not. It takes either `Some(value)` or `None`.

```
let x: Option<int> = Some(42)
print(x)          # Some(42)
```

```
let y: Option<int> = None
print(y)    # None
```

Extracting the Value

Use `match` to safely extract the inner value and handle the `None` case.

```
match x:
    case Some(v):
        print(v)    # 42
    case None:
        print("nothing")
```

Concurrency Basics

`Task<T>` is the runtime handle for concurrent work. Use `spawn` for explicit task creation, `async fn` for task-returning functions, and `await` or `join(task)` to wait for completion.

```
fn square(x: int) -> int:
    return x * x
```

```
let t: Task<int> = spawn square(12)
print(await t)    # 144
```

```
async fn add(a: int, b: int) -> int:
    return a + b
```

```
print(await add(20, 22))    # 42
await add(1, 2)              # statement form also works
```

`@parallel` can be applied to counted `for` loops over `range(...)` or integer `..` ranges:

```
@parallel
for i in range(8):
    print(i)
```

In v1, `spawn` does not support `Unit`-returning calls, and `@parallel for` rejects `break`, `continue`, and writes to outer mutable variables.

For channels, `recv(ch)` is the strict form and raises on a closed drained channel, while `recv_opt(ch)` returns `Some(value)` or `None` instead. `for x in ch:` is the close-aware consumer form and ends normally once the channel is closed and drained. For `Channel<Unit>`, `recv_opt(ch)` returns `bool` and `for _ in ch:` can be used to consume values.

F-String (String Interpolation)

Use `f"..."` to embed expressions directly inside strings. Expressions are placed in `{}`.

```
let name = "Alice"
print(f"Hello {name}")    # Hello Alice
```

```
let x = 3
let y = 4
print(f"{x} + {y} = {x + y}")    # 3 + 4 = 7
```

Use `{{` and `}}` to include literal braces.

```
print(f"{{escaped}}")    # {escaped}
```

Type Casting (as)

Convert between types explicitly with `as`.

```
let x = 42 as float      # 42.0
let y = 3.14 as int      # 3 (truncated)
let s = 42 as str        # "42"
let b = true as int      # 1
```

Enum with Associated Data (ADT)

Enum variants can carry associated values. This lets a single enum represent a family of different shapes of data.

```
enum Shape:
    Circle(float)
    Rectangle(float, float)
    Point
```

Constructing ADT Variants

```
let c = Shape::Circle(3.14)
let r = Shape::Rectangle(4.0, 5.0)
let p = Shape::Point
```

Matching ADT Variants

Use `case` with a binding pattern to extract the associated data.

```
fn describe(s: Shape) -> str:
    match s:
        case Shape::Circle(r):
            return f"circle with radius {r}"
        case Shape::Rectangle(w, h):
            return f"rectangle {w}x{h}"
        case Shape::Point:
            return "point"

print(describe(Shape::Circle(3.14)))    # circle with radius 3.14
print(describe(Shape::Rectangle(4.0, 5.0))) # rectangle 4.0x5.0
```

Generic Enum

Enums can take type parameters, making them reusable across different payload types.

```
enum MyOption<T>:
    MySome(T)
    MyNone
```

Usage

```
let a = MyOption<int>::MySome(42)
let b: MyOption<int> = MyOption<int>::MyNone
```

```
match a:
  case MyOption::MySome(v):
    print(v)          # 42
  case MyOption::MyNone:
    print("none")
```

Result Type

`Result<T, E>` is used for functions that may fail. Return `Ok(value)` for success and `Err(error)` for failure.

```
fn divide(a: int, b: int) -> Result<int, str>:
  if b == 0:
    return Err("division by zero")
  return Ok(a // b)
```

Use `match` to handle the result.

```
let r = divide(10, 0)
match r:
  case Ok(v):
    print(v)
  case Err(e):
    print(e)    # division by zero
```

[<- Prev: Collections](#) | [Next: Packages ->](#)

[English](#) | |

Packages

[<- Prev: Advanced Features](#) | [Next: Design by Contract ->](#)

Ry uses a package system to organize code across files and directories. For the full specification, see [Package Reference](#).

from/import Syntax

Use the `from` syntax to import functions from another file.

```
from math import add, sub    # Selective import
from math                    # Full import (all definitions)
```

This makes functions defined in `math.ry` available for use.

Subdirectories (Dot-Separated Paths)

Use dot-separated paths to specify packages in subdirectories.

```
from utils.math import add    # Import from utils/math.ry
```

Each dot corresponds to a directory separator.

Directory Packages

A package can be either a single `.ry` file or a directory containing multiple `.ry` files. When a package resolves to a directory, all `.ry` files in it are automatically loaded.

```
mypackage/  
  math.ry      # fn add(), fn sub()  
  string.ry    # fn concat()  
  
from mypackage      # imports add, sub, concat  
from mypackage import add  # imports only add
```

No special entry file (like `__init__.py`) is needed. Files starting with `_` are excluded.

Standard Library (std)

The `std` package is automatically imported into every program. You don't need to write `from std` — it's always available.

```
# These functions are available without any import  
print("hello")  
let n = len("world")  
let xs = range(5)
```

You can also explicitly import specific definitions from `std` sub-packages:

```
from std.str import contains
```

RY_HOME

The standard library is installed at `$RY_HOME/lib/std/`. The default value of `RY_HOME` is `~/.ry`.

```
export RY_HOME="$HOME/.ry"    # default
```

Search Path Priority

Package files are searched in the following order:

1. **Directory of the importing file** — The directory containing the file with the import statement is searched first.
 2. **`$RY_HOME/lib`** — Standard library location.
 3. **Executable-relative `lib/`** — Directories relative to the `ry` executable.
 4. **`RY_PATH` environment variable** — If not found, directories specified in `RY_PATH` are searched in order.
-

RY_PATH Environment Variable

Multiple directories can be specified separated by colons.

```
export RY_PATH=/home/user/ry-libs:/usr/local/ry-libs
```

Once set, packages in the specified directories can be imported from anywhere.

Limitations

- **from** statements can only be written at the **top level** of a file. They cannot be placed inside functions or blocks.
- Importing the same package multiple times is automatically skipped (no duplicate imports).
- **Circular imports** (A imports B and B imports A) result in an error.

```
# Error example: a.ry and b.ry import each other  
# a.ry: from b import foo  
# b.ry: from a import bar <- circular import error
```

[<- Prev: Advanced Features](#) | [Next: Design by Contract ->](#)

[English](#) | |

Design by Contract

[<- Prev: Packages](#) | [Next: Testing ->](#)

Ry supports Eiffel-style Design by Contract with preconditions (**require**), postconditions (**ensure**), and struct invariants (**invariant**). Contract violations terminate the program. For the full specification, see [Design by Contract Reference](#).

Preconditions (**require**)

Use **require** to specify conditions that must be true when a function is called.

```
fn deposit(amount: int, balance: int) -> int:  
  require:  
    amount > 0  
    balance >= 0  
  var new_balance: int = balance + amount  
  return new_balance
```

If any precondition fails, the program terminates with:

```
Contract violation: require failed in deposit()
```

Postconditions (**ensure**)

Use **ensure** to specify conditions that must be true when a function returns.

The **result** Keyword

Inside an **ensure** block, **result** refers to the return value.

```
fn abs(x: int) -> int:  
  ensure:  
    result >= 0  
  if x < 0:
```

```
    return -x
  return x
```

The old() Expression

`old(expr)` captures the value of an expression at function entry. This is useful for comparing pre- and post-states.

```
fn increment(x: int) -> int:
  ensure:
    result == old(x) + 1
  return x + 1
```

Combining require and ensure

Both can be used together. `require` must come before `ensure`.

```
fn deposit(amount: int, balance: int) -> int:
  require:
    amount > 0
    balance >= 0
  ensure:
    result >= 0
    result == old(balance) + amount
  var new_balance: int = balance + amount
  return new_balance
```

Struct Invariants (invariant)

Use `invariant` to specify conditions that must always hold for a struct. Invariants are checked after construction and after every field assignment.

```
record BankAccount:
  balance: int
  min_balance: int
  invariant:
    balance >= min_balance

let a = BankAccount(100, 0)    # OK: 100 >= 0
# a.balance = -1              # Contract violation: invariant failed
```

Rules

- `require` and `ensure` blocks are optional and appear before the function body.
 - `require` must come before `ensure` when both are present.
 - `result` and `old()` can only be used inside `ensure` blocks.
 - `invariant` appears at the end of a `record` definition, after all field declarations.
 - All contract violations terminate with `exit(1)`.
-

[<- Prev: Packages](#) | [Next: Testing ->](#)

[English](#) | |

Testing

[<- Prev: Design by Contract](#)

Ry has a built-in RSpec-style test syntax using `describe`, `it`, and `expect`. For the full specification, see [Testing Reference](#).

Running Tests

```
ry test # Auto-discover and run all *.test.ry files
ry test tests/my_test.test.ry # Run a specific test file
```

The exit code is 0 if all tests pass, 1 if any test fails.

When run without arguments, `ry test` searches for `ry.toml` to find the project root, then recursively discovers all `*.test.ry` files.

Writing Tests

Use `describe` to group related tests and `it` to define individual test cases.

```
describe("Calculator", fn():
  it("adds integers", fn():
    expect(1 + 2).to_eq(3)

  )
  it("subtracts integers", fn():
    expect(5 - 3).to_eq(2)

  )
  it("checks booleans", fn():
    expect(3 > 1).to_be_true()
  )
)
```

- `describe` and `it` take a description string and a **lambda argument** `fn():` as the second parameter
 - `describe`, `it`, `expect`, `mock`, and `verify` are only available with `ry test` (compile error with normal `ry` execution)
-

Matchers

Matcher	Description	Supported Types
<code>to_eq(expected)</code>	Equality comparison	int, float, bool, str
<code>to_not_eq(expected)</code>	Asserts not equal	int, float, bool, str
<code>to_be_true()</code>	Asserts <code>true</code>	bool
<code>to_be_false()</code>	Asserts <code>false</code>	bool
<code>to_be_none()</code>	Asserts <code>None</code>	Option
<code>to_be_some()</code>	Asserts Option is <code>Some</code>	Option
<code>to_contain(val)</code>	Asserts container includes value	List, Set, str

Output Format

Calculator

- + adds integers
- + subtracts integers
- checks booleans
- line 10: expected true, got false

2 passed, 1 failed

+ indicates pass (green), - indicates failure (red).

Mocking

`mock(fn_name, replacement)`

Replaces a function with a mock implementation for the current `it` block. The mock is automatically restored when the `it` block ends.

```
fn fetch_data() -> str:
    return "real data"
```

```
describe("mocking", fn():
    it("replaces function", fn():
        mock(fetch_data, fn(): "fake")
        expect(fetch_data()).to_eq("fake")
    )
    it("auto-restores", fn():
        expect(fetch_data()).to_eq("real data")
    )
)
```

`verify(fn_name)`

Returns the number of times a mocked function was called.

```
describe("verify", fn():
    it("counts calls", fn():
        mock(fetch_data, fn(): "fake")
        fetch_data()
        fetch_data()
        expect(verify(fetch_data)).to_eq(2)
    )
)
```

Parameterized Tests

Use `@each` to run the same test with multiple inputs:

```
@each([(1, 2, 3), (0, 0, 0), (-1, 1, 0)])
it("adds {0} + {1} = {2}", fn(a: int, b: int, expected: int):
    expect(a + b).to_eq(expected)
)
```

Each tuple becomes a separate test case. {0}, {1}, etc. in the description are replaced with actual values.

Property-Based Tests

Use `@property` to test with randomly generated inputs:

```
@property(count=100)
it("addition is commutative", fn(a: int, b: int):
    expect(a + b).to_eq(b + a)
)
```

The test runs `count` times with random values. On failure, the counterexample is printed.

Limitations

- Nesting of `describe` is not supported
 - `before_each` / `after_each` are not supported
 - Overloaded and `@native` functions cannot be mocked
-

[<- Prev: Design by Contract](#)